

Dilangi Thilakarathna

LinkedIn: <https://www.linkedin.com/in/dilangi-thilakarathne>

Email: dilangithilakarathne@gmail.com

Portfolio: <https://dilangi.github.io/>

Contact: +447928282428

Profile

Software Engineer | Java & AWS | Bristol, UK

Graduate visa valid to Feb 2027 | Full-time, no restrictions | Hybrid or remote

Software engineer with **3+ years of professional experience** across fintech, e-commerce, and life sciences. Built and shipped real software: an Android delivery app used by 100+ drivers, Spring Boot services deployed on AWS, and CI/CD pipeline improvements at a UK SaaS company. Currently building a multi-service inventory management system in Java/Spring Boot using microservices architecture. **MSc Computing (Distinction, University of Northampton, 2024)** and **AWS Certified Cloud Practitioner**. Confident in Java/Spring Boot backends, AWS infrastructure, and CI/CD pipelines. **Use AI coding tools** (Codex, Copilot, Claude) day-to-day **alongside solid fundamentals**.

Key Skills

- **Languages:** Java, JavaScript / TypeScript, Python, Kotlin , SQL
- **Backend:** Spring Boot, Spring Security, REST APIs, JWT authentication, Microservices, Node.js
- **Cloud & AWS:** EC2, RDS, S3, DynamoDB, CloudFront, Lambda, VPC, AWS Certified Cloud Practitioner
- **CI/CD & IaC:** Jenkins (Groovy DSL), Terraform, GitHub Actions, Docker
- **Testing:** JUnit, Mockito, Jest, unit & integration testing
- **Databases:** MySQL, Snowflake, DynamoDB, Firebase Firestore
- **Mobile:** Android native (Java/Kotlin), Google Play Store deployment
- **Tools:** Git, Jira, Confluence, Agile/Scrum, code review & PR workflows
- **AI tooling:** OpenAI Codex, GitHub Copilot, Claude, used to accelerate development
- **Practices:** OOP, Design Patterns, TDD, Agile, Problem-solving, Code Review

Professional Experience

Software Engineer (Placement) | [IDBS, UK](#) | Sept 2023 – Aug 2024

IDBS is a UK SaaS company providing cloud data platforms to 18 of the top 20 global biopharmaceutical companies. Agile squad within the BioPharmaceutical Lifecycle Management platform, backend services, AWS infrastructure, and CI/CD tooling.

- Refactored redundant Jenkins CI/CD jobs into a single parameterised pipeline (Groovy DSL), simplifying multi-environment deployments across dev, staging, and production
- Improved Snowflake dashboard query performance by designing row access policies that removed inner join bottlenecks, load times roughly halved for 3 of 5 views
- Upgraded TLS version on AWS CloudFront distributions via Terraform, resolving a vulnerability identified during penetration testing
- Raised JUnit/Mockito unit test coverage in a Spring Boot repository from ~45% to ~75%, writing tests for REST controller routing, JSON serialisation, CORS headers, and annotation behaviour
- Deployed a DynamoDB migration script across dev, staging, and production; diagnosed production failures and coordinated backup table cleanup with the SRE team
- Contributed to Tableau app versioning: pushed artefacts to AWS S3 and Systems Manager Parameter Store via Jenkins pipelines
- Active in all Scrum ceremonies: sprint planning, backlog refinement, story pointing, retrospectives, daily standups
- Presented intern reflections at the monthly engineering townhall, first intern to do so at IDBS

Software Engineer | [IFS R&D International, Sri Lanka](#) | Mar 2022 – Sept 2022

Global ERP software company. Short contract role.

- Bug fixes and minor features in ERP product modules using IFS's proprietary framework
- Database query work and field-level validations across the full software development lifecycle

Software Engineer | [Kapruka.com, Sri Lanka](#) | Dec 2020 – Mar 2022

Sri Lanka's largest e-commerce platform. Sole developer for an internal Android delivery management app built from scratch.

- Built the app end-to-end: card-based delivery order UI, Google Maps navigation, offline storage, image capture, digital signature pad, and Firebase Cloud Messaging push notifications for 100+ drivers
- Deployed to Google Play Store; managed the full lifecycle from requirements through production
- Implemented route sequencing feature, replacing the previous paper-card system

Associate Software Engineer | [Epic Lanka, Sri Lanka](#) | Mar 2019 – Sept 2020

Fintech mobile development — branchless banking and QR payment apps.

- Built Android UI screens for banking and QR payment applications, integrating backend APIs
- Developed professional coding and version control practices within a senior-led engineering team

Technical Projects

[Spare Parts Inventory Management System](#) (2026) | in active development

Multi-service backend for a real hardware retail business, replacing manual stock and sales tracking with a structured digital system across three branches. MVP include:

- Microservices architecture in Java / Spring Boot: Auth, Inventory, Sales, and Supplier services, each with an independent database schema
- Auth service: JWT token generation, BCrypt password hashing, role-based access control (Developer / Owner / Manager / Operator)
- Inventory service: product management, branch-level stock tracking, pricing, categories, brands, and product condition (new / reconditioned)
- Sales service: invoice generation with branch-prefixed numbering, line-item and invoice-level discounts, exclusive tax calculation, automatic stock reduction
- Used OpenAI Codex to accelerate development; responsible for all architecture decisions, service design, and backend implementation

[Spring Boot Microservices API with Authentication](#) (2025) | Java · Spring Boot · AWS EC2/RDS · JWT

- REST API with JWT authentication, BCrypt password hashing, CRUD operations, input validation, and JUnit/Mockito unit tests
- Deployed to AWS EC2 with RDS MySQL and VPC security configuration

[IoT Smart Door System](#) (MSc Project, 2023) | Arduino · Python · Android · Bluetooth

- IoT smart door prototype with Android companion app for real-time monitoring and alerts; documented as an IEEE-format technical paper

Education and Certifications

- **MSc. Computing** (Distinction, includes 12-months placement)
University of Northampton, UK | 2022 - 2024
- **BSc. Computer Science** (2:2)
University of Colombo School of Computing, Sri Lanka | 2016 - 2019
- [AWS Certified Cloud Practitioner](#) (Oct 2025)